

JOB PORTAL MANAGEMENT SYSTEM

School of Computing

**Bachelor of Technology
in
Computer Science & Engineering**

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Full Stack Application Development

Slot : E3-B9



Figure 1:

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May, 2026

BONAFIDE CERTIFICATE

It is certified that the work contained in the Full Stack Application Development report titled "Job Portal Management System" by Pula Likitha (VTU28727) has been carried out in Winter semester of Academic year 2025-2026(WS2526).

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DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. I also declare that i have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

Job Portal Management System is a web-based application designed to provide an efficient platform for connecting job seekers and recruiters. The system enables users to register, log in, create personal profiles, upload resumes, search available jobs, and apply for suitable job opportunities online. It also allows administrators or employers to add job vacancies, manage job postings, and monitor candidate applications. The project is developed using Java, Spring Boot, Thymeleaf, HTML, CSS, and MySQL database. It offers a user-friendly interface that simplifies the recruitment process and reduces manual work involved in traditional hiring methods. The main objective of this system is to automate job searching and recruitment activities by creating a centralized online platform. It helps job seekers easily access opportunities and allows recruiters to efficiently manage applicants. The Job Portal Management System provides a secure, reliable, and scalable solution for modern employment management.

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LIST OF ACRONYMS AND ABBREVIATIONS

CI	Continuous Integration
ORM	Object Relational Mapping

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Chapter 1

INTRODUCTION

1.1 Introduction

The Job Portal Management System is a web-based application developed to simplify the process of recruitment and job searching. In the modern world, many companies require a fast and efficient method to hire suitable candidates, while job seekers need an easy platform to find available opportunities. This system provides a centralized solution where employers and applicants can interact through an online portal.

The application allows users to register, log in, create profiles, upload resumes, search jobs, and apply for vacancies based on their qualifications and interests. Administrators or recruiters can post new jobs, manage job listings, and review applications submitted by candidates. This reduces paperwork, saves time, and increases the speed of communication between employers and applicants.

The project is developed using Java, Spring Boot, Thymeleaf, HTML, CSS, and MySQL database. These technologies help in building a secure, scalable, and user-friendly system. The portal is designed with separate modules for users and administrators to ensure proper management and functionality.

The main purpose of the Job Portal Management System is to digitize the traditional recruitment process and make job searching more convenient, transparent, and effective. It serves as a modern solution for employment management in today's competitive world.

1.2 Aim of the Project

The aim of the Job Portal Management System is to develop an online platform that simplifies the recruitment process by connecting job seekers and employers in a single system. It is designed to help users search and apply for jobs easily, while enabling administrators or recruiters to post vacancies, manage job listings, and monitor applications efficiently. The project aims to reduce manual work, save time, improve communication, and provide a secure and user-friendly solution for modern employment management.

1.3 Scope of the Project

The scope of the Job Portal Management System is to provide a complete online recruitment platform for job seekers and employers. The system allows users to register, create profiles, upload resumes, search for jobs, and apply for suitable opportunities from anywhere at any time. It also enables administrators or recruiters to post job vacancies, update job details, view applicants, and manage the hiring process efficiently.

The project can be used by companies, colleges, consultancies, and employment agencies to streamline recruitment activities. It reduces paperwork, saves time, and improves accessibility through a centralized web-based system. In the future, the system can be enhanced with features such as email notifications, interview scheduling, skill-based matching, analytics dashboards, and mobile application support.

1.4 Framework (Spring Boot)

The Job Portal Management System is developed using **Spring Boot**, a Java-based backend framework used to build fast and scalable web applications.

It follows the **MVC architecture**, where:

- **Model** handles data (users, jobs, applications)

- **View** represents the user interface
- **Controller** manages requests and business logic

Spring Boot uses Spring MVC for request handling and Spring Data JPA for database connectivity (MySQL). It also provides REST APIs for communication between frontend and backend and runs on an embedded Tomcat server.

Advantages:

- Reduces configuration complexity
- Improves development speed
- Supports scalability and security
- Efficient backend processing

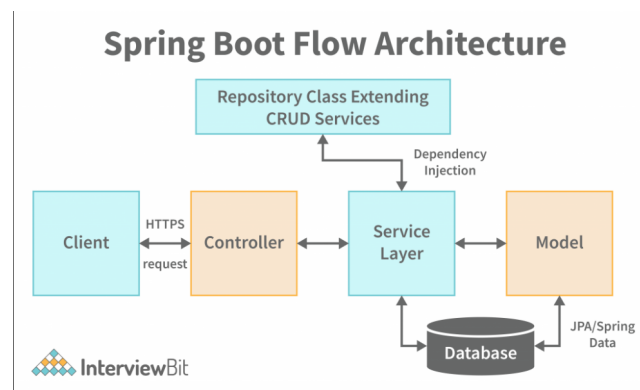


Figure 1.1: Spring Boot Framework is a popular Java-based framework used to build standalone, production-ready Spring applications with minimal configuration

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

A number of job portal applications are already available in the market that help job seekers find employment opportunities and assist recruiters in hiring suitable candidates. These applications provide various features such as job search, resume management, and applicant tracking. A study of some popular job portals is given below:

2.1 LinkedIn

LinkedIn is a professional networking platform that also functions as a job portal. It allows users to create professional profiles, upload resumes, and apply for jobs directly.

Features:

- Professional profile creation
- Job recommendations using algorithms

- Direct communication with recruiters
- Networking with professionals

Advantages:

- Large professional network
- Global job opportunities
- Real-time recruiter interaction

Limitations:

- Many advanced features are paid
- High competition among applicants

2.2 Indeed

Indeed is one of the most widely used job search engines that aggregates job listings from various sources.

Features:

- Job search by title, location, and category
- Resume upload facility
- Company reviews and salary insights

Advantages:

- Large number of job listings
- Simple and user-friendly interface
- Free for job seekers

Limitations:

- Duplicate job postings
- Limited direct recruiter interaction

2.3 Naukri.com

Naukri.com is one of the leading job portals in India providing employment opportunities across various sectors.

Features:

- Resume database for recruiters
- Job alerts via email and SMS
- Advanced search filters

Advantages:

- Strong presence in India
- Large recruiter base
- Regular job updates

Limitations:

- Some features are paid
- Interface may be complex for beginners

2.4 Glassdoor

Glassdoor is a job portal that also provides company reviews, salary details, and interview experiences.

Features:

- Job listings
- Company reviews
- Salary and interview insights

Advantages:

- Transparent company information
- Helps in better job decision-making

Limitations:

- Fewer job listings compared to competitors
- More focused on reviews than recruitment

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend of the Job Portal Management System is developed using basic web technologies such as HTML, CSS, and JavaScript. A code editor like Visual Studio Code is used for development. The project runs on a web browser such as Chrome or Edge without any additional installation.

3.1.2 Structure of the Project

The frontend project is organized into different files and folders for better management:

- index.html – Main landing page
- login.html / register.html – User authentication pages
- dashboard.html – User dashboard

- `css/` – Styling files
- `js/` – JavaScript files for interactivity
- `images/` – Used for UI design assets

3.1.3 Execution Procedure

To run the frontend:

- Open the project folder in VS Code
- Open **index.html** file
- Run using Live Server or open directly in a browser
- Navigate through pages like login, registration, and dashboard

3.2 Backend Implementation

3.2.1 Environment Setup

The backend is developed using Java (Spring Boot / Servlet-based system) with MySQL database for data storage. Tools used include:

- Eclipse / IntelliJ IDEA / VS Code
- JDK installed
- MySQL Server / XAMPP
- Maven (if Spring Boot is used)

3.2.2 Structure of the Project

The backend follows a layered structure:

- Controller Layer – Handles user requests
- Service Layer – Contains business logic
- Repository Layer – Communicates with database
- Model Layer – Defines database entities like User, Job, Application
- Database (MySQL) – Stores all user and job data

3.2.3 Execution Procedure

To run the backend:

- Start MySQL server
- Import database schema
- Run backend application from IDE
- Start server (Tomcat / Spring Boot)
- Connect frontend with backend APIs
- Access system via browser

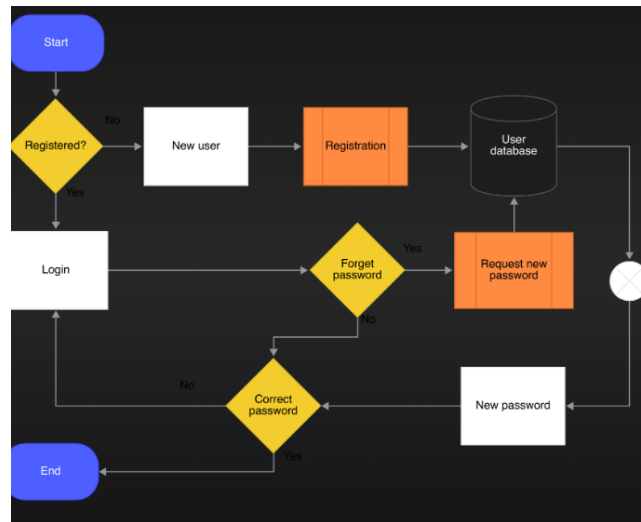


Figure 3.1: The execution procedure describes the step-by-step process followed to run and test the application successfully.

3.3 Database Implementation

The database is used to store and manage all the data related to users, jobs, applications, and recruiters in the Job Portal Management System. It ensures data security, consistency, and easy retrieval.

3.3.1 Database Configuration

The database is implemented using MySQL. The system is connected to the backend using database connectivity (JDBC / Spring Data JPA depending on project type).

- **Database name:** job_portal
- **Host:** localhost
- **Port:** 3306

- **Username:** root
- **Password:** user

The backend application establishes a connection with the database to perform operations like insert, update, delete, and retrieve data.

3.3.2 Schema Structure

The database contains multiple tables to manage different modules of the system:

- **User Table**

- user_id (Primary Key)
- name
- email
- password
- role (Job Seeker / Recruiter)

- **Job Table**

- job_id (Primary Key)
- title
- description
- company

- location
- salary

- **Application Table**

- application_id (Primary Key)
- user_id (Foreign Key)
- job_id (Foreign Key)
- status (Applied / Selected / Rejected)

- **Recruiter Table**

- recruiter_id (Primary Key)
- company_name
- email

These tables are linked using primary and foreign keys to maintain relationships between data.

3.3.3 Tables Created

The following tables have been created in the database to manage different functionalities of the Job Portal Management System:

- **User Table** – Stores details of job seekers and recruiters such as name, email, password, and role.

- **Job Table** – Stores job-related information including title, description, company, location, and salary.
- **Application Table** – Maintains records of job applications with user ID, job ID, and application status (Applied/Selected/Rejected).
- **Recruiter Table** – Stores recruiter details such as company name and email.

These tables are properly linked using primary keys and foreign keys, which helps in maintaining data consistency and relationships between different modules of the system.

Chapter 4

METHODOLOGIES

4.1 Project Architecture

The Job Portal Management System follows a 3-tier architecture consisting of:

- **Frontend Layer** – User interface developed using HTML, CSS, and JavaScript.
- **Backend Layer** – Handles business logic using Java / Spring Boot.
- **Database Layer** – MySQL database used to store and manage data.

This architecture ensures separation of concerns, making the system scalable, maintainable, and efficient.

4.2 Data Flow Diagram

The Data Flow Diagram (DFD) of the Job Portal Management System represents the flow of data between users, processes, and the database. It shows how information is processed in the system from input to output.

4.2.1 Level 0 DFD (Context Diagram)

At the highest level, the system is represented as a single process called **Job Portal System**.

- **Job Seeker** provides details such as registration, login, and job application.
- **Recruiter** provides job postings and manages applications.
- The system processes all inputs and interacts with the database.
- Output includes job listings, application status, and user information.

4.2.2 Level 1 DFD

The system is divided into main processes:

- **User Management Process** – Handles registration, login, and profile management.

- **Job Management Process** – Allows recruiters to post jobs and job seekers to view jobs.
- **Application Management Process** – Handles job applications and updates application status.
- All processes interact with the **Database** to store and retrieve data.

4.2.3 Data Flow Description

- Data flows from users (Job Seekers and Recruiters) into the system.
- The system processes requests like login, job search, and application submission.
- The database stores user data, job details, and application records.
- Processed data is sent back as output such as job listings and application status updates.

4.2.4 Conclusion

The Data Flow Diagram helps in understanding the working of the system clearly by showing how data moves between different modules. It ensures proper design and smooth functioning of the Job Portal Management System.

4.3 Module 1: User Management

This module handles all user-related operations.

- User registration (Job Seeker / Recruiter)
- Login and authentication
- Profile management
- Resume upload (for job seekers)

This module ensures secure access and proper user identity management.

4.4 Module 2: Job Management

This module manages job-related operations.

- Recruiter can post new jobs
- Job seekers can view and search jobs
- Job details include title, description, company, location, and salary
- Users can apply for jobs

This module connects recruiters and job seekers effectively.

4.5 Module 3: Application Management

This module handles the job application process between job seekers and recruiters.

- Job seekers can apply for available jobs
- Applications are stored in the database
- Recruiters can view and manage applications
- Application status is updated as Applied / Selected / Rejected

This module ensures smooth communication between users and recruiters during the hiring process.

4.6 Advantages of this Project

- **Easy Job Search:** The system allows users to easily search and apply for jobs based on their skills and location.
- **Efficient Recruitment Process:** Recruiters can quickly post jobs and manage applications in a structured manner.
- **User-Friendly System:** The interface is simple and easy to use for both job seekers and recruiters.

4.7 Disadvantages

- Requires internet connection to access the system
- Limited to digital users (not suitable for offline use)
- Depends on database security for protecting user data
- May face performance issues with large-scale data if not optimized

Chapter 5

RESULTS AND DISCUSSIONS

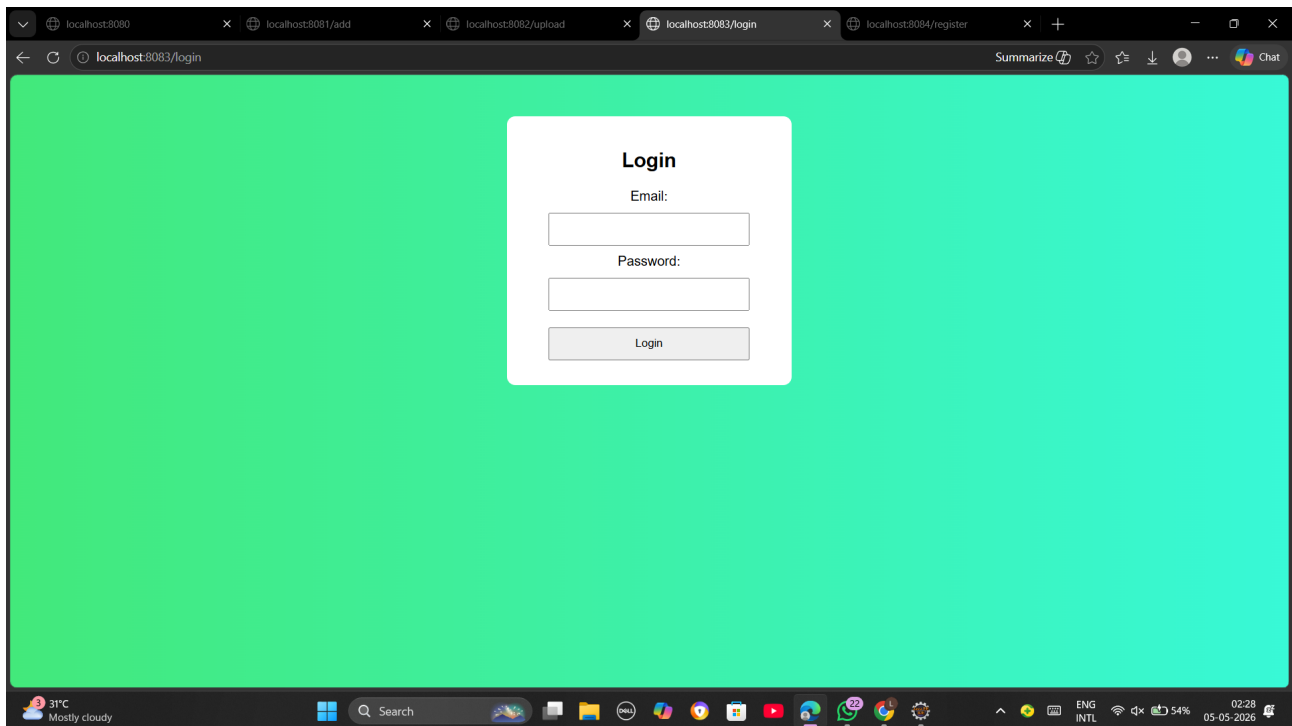


Figure 5.1: The Login Page in the Job Portal allows registered users to securely access their accounts by validating their credentials and directing them to their respective dashboards.

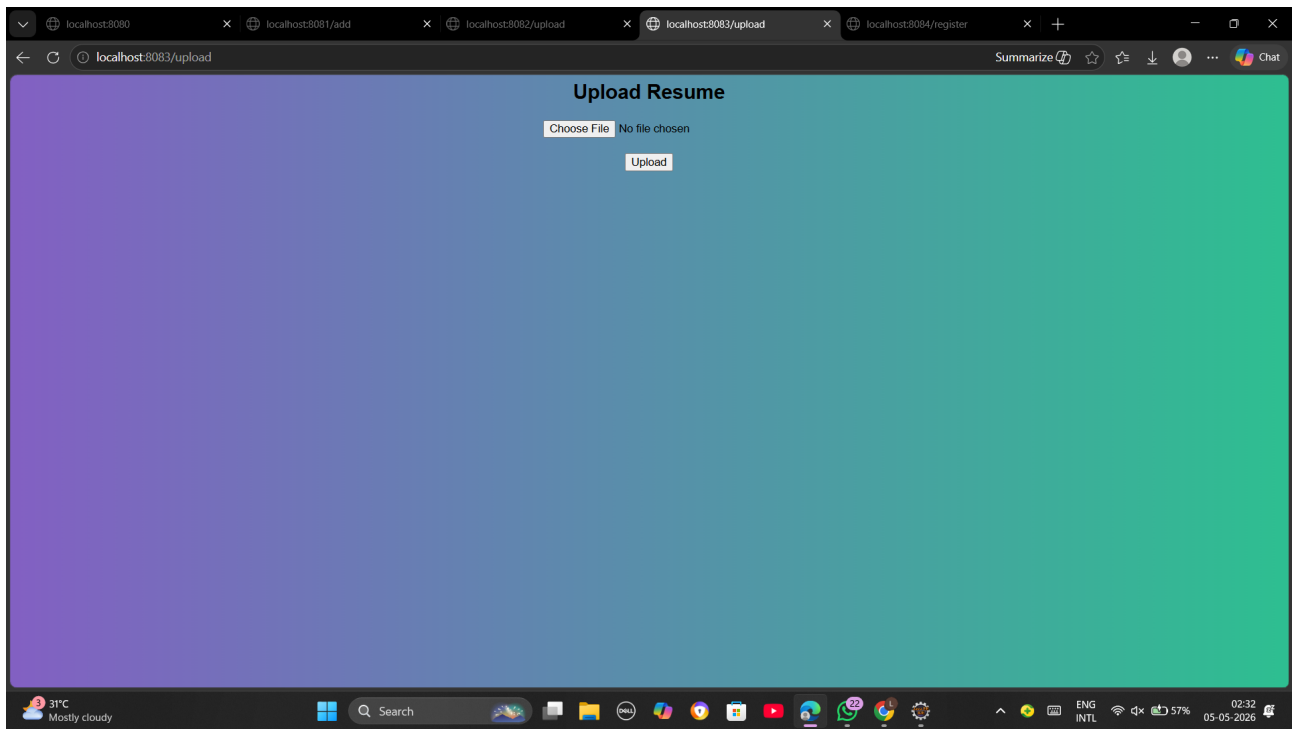


Figure 5.2: The Resume Upload and Profile Details section allows job seekers to upload their resumes and enter personal, educational, and skill information to create a complete professional profile in the job portal.

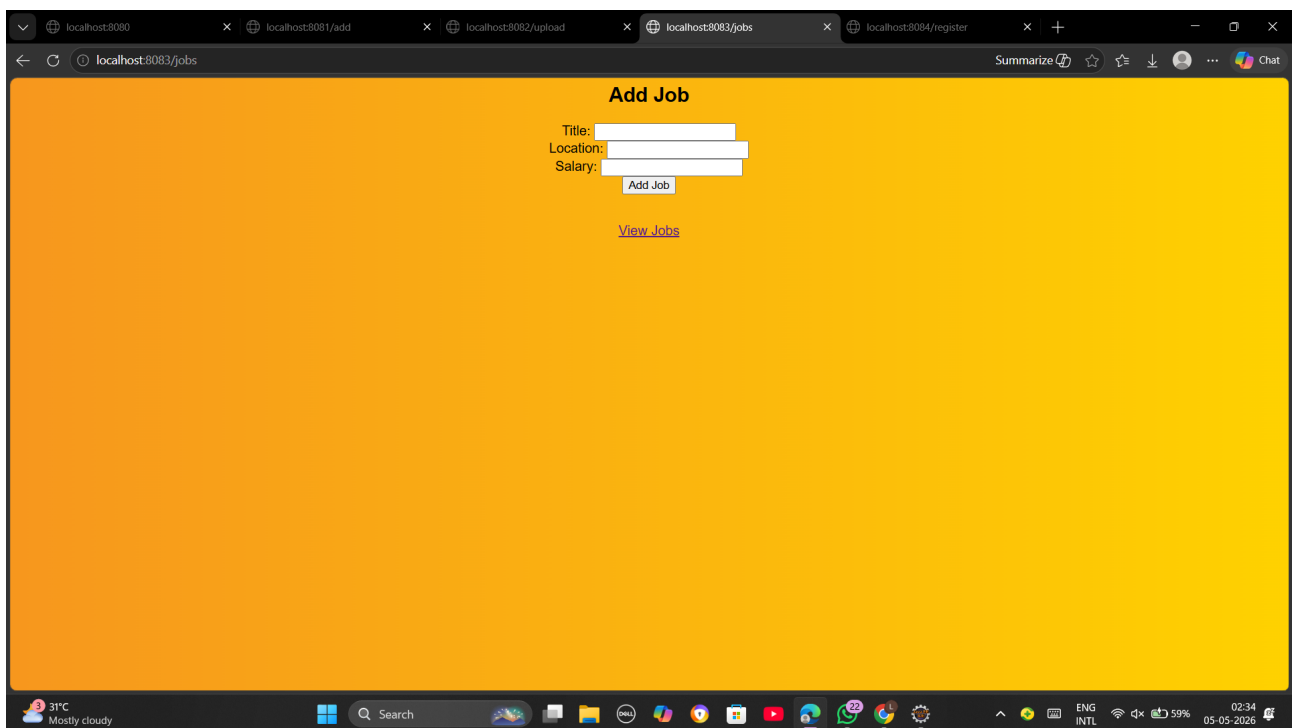


Figure 5.3: The Add Jobs feature allows recruiters to post new job openings by entering job details such as title, description, skills required, and salary, making them available for job seekers to apply.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Job Portal Management System is successfully designed to simplify the process of job searching and recruitment. It provides an efficient platform where job seekers can easily find and apply for jobs, while recruiters can post job openings and manage applications effectively.

The system reduces manual effort, saves time, and improves communication between employers and candidates. By using a structured database and user-friendly interface, the application ensures better data management and smooth operation.

Overall, the project achieves its main objective of developing a simple, efficient, and user-friendly job portal system that enhances the recruitment process for both job seekers and recruiters.

6.2 Future Enhancements

The Job Portal Management System can be further improved in the future by adding advanced features to enhance user experience and system efficiency:

- Integration of Artificial Intelligence (AI) for smart job recommendations based on user skills and profile.
- Implementation of chat system between recruiters and job seekers for direct communication.
- Addition of video interview scheduling feature within the platform.
- Mobile application development for better accessibility on smartphones.
- Advanced filtering and sorting options for faster job search.
- Integration with external job APIs for wider job listings.

These enhancements will make the system more modern, scalable, and competitive with existing job portals.

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Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	Wks	Justification
WP1	Depth of Knowledge	In-depth engineering knowledge	WK3, WK4, WK5, WK6	Knowledge of Spring Boot, MVC, Security, JPA, and DB design for job portal
WP2	Conflicting Requirements	Technical & non-technical conflicts	WK4, WK5, WK6	Balances security and usability for different user roles
WP3	Depth of Analysis	Analytical thinking	WK3, WK4, WK5	ER modeling, API design, file upload, email integration
WP4	Familiarity	Partially new problems	WK4, WK8	Job tracking introduces challenges beyond CRUD
WP5	Applicable Codes	Limited standard solutions	WK6	Custom design for resume storage and filtering
WP6	Stakeholder Needs	Multiple stakeholders	WK5, WK6, WK7	Supports job seekers, employers, admins
WP7	Interdependence	System integration	WK3, WK5, WK6	Integrates frontend, backend, DB, security, email

Table 7.1: Mapping of Project to Complex Engineering Problem (CEP)

7.2 Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification
EA1	Range of Resources	Use of diverse technologies	Uses Spring Boot, Spring MVC, JPA, MySQL/H2, Thymeleaf, Spring Security, JavaMailSender
EA2	Level of Interactions	Complex interactions	Manages job listings, applications, authentication, resume uploads, and notifications
EA3	Innovation	Modern technologies	Role-based dashboards, dynamic filtering, file upload, email notifications
EA4	Consequences	Impact on society	Improves employment accessibility and reduces manual hiring work
EA5	Familiarity	Beyond standard practices	Combines authentication, tracking, resume handling, and dashboards

Table 7.2: Mapping of Project to Complex Engineering Activities (CEA)

7.3 SDG Mapping (CEP Section)

SDG No.	SDG Title	Relevance to the Project
SDG 8	Decent Work and Economic Growth	Connects job seekers with employers and improves career opportunities
SDG 9	Industry, Innovation and Infrastructure	Promotes digital recruitment using APIs and automation
SDG 10	Reduced Inequalities	Provides equal access regardless of location
SDG 16	Peace, Justice and Strong Institutions	Ensures security using role-based access and encryption

Table 7.3: SDG Mapping for the Project